

Ritabrata Chakraborty

[GitHub](#) | [Google Scholar](#) | [Portfolio](#) | [LinkedIn](#) | ritabratabits@gmail.com | +91 89107 83548

EDUCATION

University of Pennsylvania

M.S.E. in Robotics, GPA: -/4

Pennsylvania, USA

Aug '26 – May '28

Birla Institute of Technology and Science, Pilani (BITS Pilani)

B.E. in Mechanical Engineering — Minor in Data Science, CGPA: 8.57/10

Rajasthan, India

Oct '22 – May '26

- **Institute Merit-cum-Need Scholarship** — Consecutive Recipient (Top 8% of Students)

LEADERSHIP & VOLUNTEERING

Deputy Director of Academic Programming

Graduate and Professional Student Assembly (GAPSA), University of Pennsylvania

Aug '26 – Present

Vice-President

South Asian Association (Rangoli), University of Pennsylvania

Aug '26 – Present

President & Secretary

Mechanical Engineering Association (MEA), BITS Pilani

Jun '24 – May '26

- Coordinated 10+ events and career sessions for 300+ students, facilitating technical exposure and alumni interaction.
- Managed production of 500+ merchandise items and led outreach, boosting student participation by 20%.

President & Tech Fest Coordinator

Indian Society of Heating, Refrigerating, and Air Conditioning Engineers (ISHRAE), BITS Pilani

Oct '24 – Jul '25

- Led a team of 25+ members to organize 4+ technical workshops with HVAC industry experts.
- Hosted 3 competitions and networking events, engaging 200+ students in HVAC innovation and awareness.

Project Manager

Tinkerer's Lab (TL), BITS Pilani

May '24 – Jul '25

- Supervised 5 interdisciplinary robotics teams (30+ members) on Micromouse and Hexapod projects.
- Oversaw lab resources, conducted weekly reviews, and mentored 50+ students in hands-on technical skills.

Treasurer

Bengali Cultural Association (Moruchhaya), BITS Pilani

May '24 – Jul '25

- Allocated funds and maintained financial records for merchandise sales, cultural programs, and annual events.

Volunteer

Junoon, National Service Scheme (NSS), BITS Pilani

Sep '23 – Sep '23

- Organized sports and recreational competitions for specially abled children from 25+ colleges.

TEACHING

Teaching Assistant

CIS 4190/5190: Applied Machine Learning, University of Pennsylvania

Aug '26 – Dec '26

CS F320: Foundations of Data Science, BITS Pilani

Jan '26 – May '26

ME F218: Advanced Mechanics of Solids, BITS Pilani

Jan '25 – May '25

ME F216: Materials Science and Engineering, BITS Pilani

Sep '24 – Dec '24

- Assisted 100+ students in labs and tutorials, clarifying concepts and linking theory to practical applications.
- Analyzed assignments and facilitated faculty in delivering high-impact teaching sessions.

PUBLICATIONS & PEER REVIEWS

Peer Reviews

- Selected as a peer reviewer for **IEEE Gujarat Section Conference**, 2026.

INDUSTRY EXPERIENCE

ML Intern | Uber (AI Solutions)

Supervisor: [Mr. Siddarth Malreddy](#), Tech Lead Manager & [Mr. Ishan Nigam](#), Senior ML Engineer, Uber

Jul '25 – Dec '25

- Streamlined object tracking in [uLabel](#) for RGB/IR videos, flagging object annotation anomalies via XGBoost (mAP50: 75%).
- Screened object-association errors with a [TrackTrack](#)-based cross-frame consistency check (F1: 88%).

Subject Matter Expert | Knowledge Cultivator

Self-supervised

Nov '23 – Feb '24

- Resolved numerical problems for student clients using MATLAB, CAD, and Python.
- Authored academic content across diverse research topics.

AWARDS & ACHIEVEMENTS

Finalist — Meta PyTorch OpenEnv Hackathon (India) 🔗	Apr '26
3 rd Place — MathWorks Global Drone Student Challenge 2025	Mar '25
Finalist — ISRO Immersion Challenge (AI for Space & Geospatial Innovation) 🔗	Jul '24
Top 15 Overall, Top 5 in College — American Express Campus Challenge 2024 🔗	Jul '24
5 th Place & Best Maintenance Award — European Rover Challenge 2023 (Remote Finals)	Sep '23

PROFESSIONAL ACTIVITIES

Graduate Student Member — IEEE	2026
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COMPETITIVE EXAMINATIONS

GRE: 317 (Quantitative: 168, Verbal: 149, AWA: 3)	2025
TOEFL iBT: 108 (Reading: 27, Listening: 30, Speaking: 23, Writing: 28)	2025
JEE Advanced: All India Rank 14372	2022

SKILLS

Leadership & Communication: Event Planning, Team Management, Budget Allocation, Mentorship, Cross-Functional Coordination, Public Speaking, Stakeholder Outreach, Technical Writing
Programming Languages: Python, C++, C, C#, Java, Bash (Shell)
Productivity Software: Microsoft Office (Excel, PowerPoint, Word), Google Workspace